Section-2

CTs, CVTs and PTs

1. SCOPE: Refer Clause-1 of Section-I, General

2 CLIMATIC CONDITIONS: Refer Clause-2 of Section-I, General

3 GENERAL TECHNICAL REQUIREMENTS: Common for all instrument transformers (CTs, CVTs and PTs) :

3.1 Construction :

The core shall be high grade non-ageing electrical silicon laminated steel of low hysterisis loss and high permeability to ensure high accuracy at both normal and over current voltage.

The secondary terminals shall be brought into a compartment on one side of Instrument transformer for easy access. There shall be separate secondary terminal box for the dedicated metering core. The secondary terminals shall be provided with short circuiting arrangements in case of Current Transformers. The secondary taps shall be adequately reinforced to withstand normal handling without damage. The ratio taps shall be on secondary side and the sealing should be provided for the dedicated core.

The mounting of instrument transformers shall be pedestal type and shall be suitable for mounting on steel structures or concrete pedestals. The necessary flanges, bolts etc., for the base of instrument transformers shall be supplied and they shall be galvanized.

All windings shall be made of double paper covering insulation and the manufacturing of the units shall be done in completely closed and air-conditioned room, otherwise fiber glass insulation sleeves are to be provided for primary winding. Details of winding and core shall be furnished.

The Instrument Transformers shall be complete in all respects with first filling of oil conforming to IS-335/IEC60296. It shall have oil level indicator with minimum/maximum oil level marks. The top cover and terminal box cover should be such that rain water does not enter even through the "O "Rings. Gaskets are not allowed. The manufacturer of CT shall test the oil for DGA before filling inside the CT and shall keep record of the same.

3.1.1. The insulation of the Instrument transformers shall be so designed that the internal insulation shall have higher electrical withstand capability than the external insulation. The designed dielectric-withstand values of external and internal insulation shall be clearly brought out in the guaranteed technical particulars. The Guaranteed Technical Particulars as given in the Specification shall invariably be mentioned in the bids furnished.

Those bids not containing the Guaranteed Particulars will be treated as Non-Responsive and rejected. The dielectric withstand values specified in this specification are meant for fully assembled Instrument Transformers.

3.1.2 Porcelain Housing: /Polymer housing

The porcelain/Polymer housing shall be in single piece only, without any metallic joints. The housing

shall be made of homogeneous, vitreous porcelain of high mechanical and dielectric strength. Glazing of porcelain shall be of uniform brown or dark brown color free from blisters, burns etc., with a smooth surface arranged to shed away rain water or condensed water particles (fog). The profile of porcelain shall be aerodynamic type as per IEC-815. The metal flanges of cast iron cemented to porcelain of the CT shall be of HDG.

3.1.2.2 Details of attachment of metallic flanges to the porcelain bushing shall be brought out in the offer. (collar type without cementing is not acceptable)

3.1.3 The metal tanks shall have bare minimum number of welded joints so as to minimize possible locations of oil leakage. The metal tanks shall be made out of mild steel galvanised/aluminum sheet /aluminum casting /stainless steel/Only. Welding in horizontal plane is to be avoided as welding at this location may give way due to vibrations during transport resulting in oil leakage. The thickness of the metal tank shall not be less than 5mm and on sides shall not be less than 3.15 mm.

3.1.4 Surface Finish:

The metal tank if made of mild steel, can be Hot Dip Galvanized, whereas, the inner portion shall be painted with Two coats of hot oil resistive insoluble paint duly after removing the zinc layer with sand blasting.

For Aluminium tanks and stainless steel, external surface of aluminium can have natural finish. The galvanizing shall be as per applicable standard IS:2629 and minimum thickness of zinc coating shall be 610 gm/Sq.mts (86 microns).

All the ferrous hardware, exposed to atmosphere, shall be hot dip galvanized or Alu zinc coating. All other fixing nuts, bolts, washers shall be made out of stainless steel only.

3.1.5 Insulating Oil:

Uninhibited Un-used EHV grade Insulating oil required for first filling of the Instrument transformer shall be covered in Bidder's scope of supply. The oil shall meet the requirements of latest edition of IS 335/IEC 60296. **TC with invoice will be verified at the time of inspection.**

3.1.6 Prevention of Oil Leakage and Entry of Moisture:

3.1.6.1 The Bidder shall ensure that the sealing of Instrument Transformer is properly achieved. In this connection the arrangement provided by the Bidder at various locations including the following ones shall be described, supported by sectional drawings.

i) Locations of emergence of primary and secondary terminals.

ii) Interface between porcelain housing and metal tanks.

iii) Cover of the secondary terminal box.

iv) Oil resistant Epoxy cast resin/FRP bushings are to be used for primary terminals.

v) For terminating secondary connections from windings, molded oil-resistant epoxy cast resin terminal cover should be used. ”O” Rings should be as per clause 3.1.6.3

vi) Between Top tank and cover.

vii) Between top cover and steel metallic bellow.

3.1.6.2 Nuts and bolts or screws used for fixation of the interfacing porcelain bushings for taking out terminals, shall be provided on flanges cemented to the bushings and not on the porcelain/Poymer.

3.1.6.3 For joints wherever used, nitrite butyl rubber/ Viton /EPDM/Neoprene "O Rings" shall be used. The "O Rings"shall be fitted in properly machined groove with adequate space for accommodating the **"O Rings"**under compression. The Test Certificate of the "O Rings" shall be submitted to the inspector during inspection.

3.1.7 Oil Level Indicators:

For all instrument transformers, stainless steel, metal bellows shall be provided for compensation of oil volume variation. Steel diaphragms are not allowed. A ground glass window shall be provided to monitor the position of metal bellow.

For 33kV and above live tank current transformers, stainless steel metal bellows shall be provided for compensation of oil volume variation. The level of shell shall be visible with naked eye to an observer standing at ground level. For dead tank Current Transformers Nitrogen cushioning is allowed (measures taken to minimize contact of nitrogen with oil is to be clearly explained in the drawing.)

3.1.8 Earthing:

Metal tank of the Instrument Transformer shall be provided with two separate Earthing terminals for bolted connection to 50 x 8 mm MS/GI flat to be provided by the Purchaser for connection to station earth-mat. There shall not be painting on the earth terminals.

3.1.9 Instrument Transformer shall be provided with suitable lifting arrangement to lift the entire unit. The lifting arrangement shall be clearly shown in the general arrangement drawing. Lifting arrangement (lifting eye) shall be positioned in such a way as to avoid any damage to the porcelain housing or the tanks during lifting for installation/transport. Necessary removable type string guides shall be provided.

3.1.10 Name Plate:

The Instrument Transformer shall be provided with non-corrosive, legible name plate with the information specified in relevant standards, duly engraved/punched on it.

Name plate shall conform to the requirements of IEC incorporating the year of manufacture. The rated current & extended current rating in case of current transformers and rated voltage, voltage factor & intermediate voltage in case of voltage transformers shall be clearly indicated on the name plate. Warranty period shall also be mentioned on the name plate.

Name plate shall include Name of the purchaser and Purchase Order No.,Warranty period Creepage distance & IEC reference standard. The name plate shall not be fixed to the removable secondary terminal box cover plates.

3.1.11 Terminal connectors :

The terminal connectors required for connection of the Instrument transformer to Purchaser's bus bar, shall be in Bidder's scope. In case of twin/quadruple conductor the terminal connector shall be suitable for both vertical and horizontal take off positions.

1) Terminal connectors shall be suitable for single/double Moose ACSR conductor and suitable for horizontal or vertical take off positions. Confirmation from APTRANSCO shall be obtained regarding type of terminal connectors at the time of drawing approval.

2) Bushing terminals shall be provided with terminal connectors of approved type and size for connection to external parts. Terminal connectors offered must have been successfully type tested as per IS 5561 and must be procured from APTRANSCO registered vendors only.

   Type test reports for Terminal connectors shall be provided from original manufacturers

3) i) All castings shall be free from blow holes, surface blisters, cracks and cavities. All sharp edges and corners shall be blurred and rounded off.

ii) No part of a clamp shall be less than 10 mm thick.

iii) All steel bolts & nuts shall be hot dip galvanized conforming to IS 1867 part xiii and spring washers shall be Electro galvanized conforming to IS 1573.

iv) For bimetallic clamp, copper alloy linear of minimum thickness of 2 mm shall be supplied which shall fix tightly.

v) Flexible connectors shall be made out of tinned copper sheets.

vi) Size of terminal/conductor for which the clamp is suitable and rated current under site conditions shall be embossed/punched on each component of the clamp, except hardware.

vii) All current carrying parts shall be designed and manufactured to have minimum contact resistance.

viii) Clamps shall be designed corona controlled.

ix) The short time rating of terminal connector shall correspond to short time rating of respective bushing.

All power clamps and connectors shall conform to IS: 5561:2018 & NEMA CC1 and shall be made of materials listed below:

a	For connecting ACSR conductors	Aluminum alloy casting, conforming to designation 4600 (A6) of IS: 617 and all test shall conform to IS : 617
b	For connecting equipment terminals made of copper with ACSR conductors	Bimetallic connectors made from aluminum alloy casting, conforming to designation 6 of IS 617 with 2mm thick bimetallic liner and all test shall conform to IS:617
c	For connecting G.I wire	Galvanized mild steel shield
d	i) bolts, nuts & Plain, washers	i) Electrogalvanized for sizes below M12, for others hot dip galvanized.
	ii) Spring washers for items ‘a’ to ‘c’	ii) Electrogalvanized mild steel suitable for at-least service condition-3 as per IS: 1573

Each equipment shall be supplied with the necessary terminals and connectors, as required by the ultimate design for the particular installation. The requirement regarding external corona and RIV as specified for any equipment shall include its terminal fittings and the equipment shall be factory tested with the connectors in position. If corona rings are required to meet these requirements they shall be considered as part of that equipment and included in the scope of work.

Where copper to aluminum connections are required, Bi-metallic clamps shall be used, which shall be properly designed to ensure that any deterioration of the connection is kept to a minimum and restricted to parts which are not current carrying or subjected to stress. The design details of the joint shall be furnished to the Purchaser by the Contractor.

Low voltage connectors, grounding connectors and accessories for grounding all equipment as

specified in each particular case, are also included in the scope of work.

No current carrying part of any clamp shall be less than 10 mm thick. All ferrous parts shall be hot dip galvanized. Copper alloy liner of minimum 2 mm thickness shall be cast integral with aluminum body for Bi-metallic clamps.

All casting shall be free from blow holes, surface blisters, cracks and cavities. All sharp edges and corners shall be blurred and rounded off.

Clamp shall be designed to carry the same current as the conductor and the temperature rise shall be equal or less than that of the conductor at the specified ambient temperature. The rated current for which the clamp/connector is designed with respect to the specified reference ambient temperature, shall also be indelibly marked on each component of the clamp / connector, except on the hardware.

All current carrying parts shall be designed and manufactured to have minimum contact resistance.

Clamps and connectors shall be designed to be corona controlled for 220kV & 132kV class.

Corona extinction voltage for 220kV & 132kV class clamps shall not be less than 156kVrms, 105kVrms respectively and R.I.V level shall not be more than 1000 micro volts at the test voltage specified in respective sections. Buffing of clamp/ connector shall be done for 132kV & above class.

TESTS

Clamps and connectors should be type tested as per IS: 5561:2018 as detailed below:

Temperature rise test (maximum temperature rise allowed is 35 degree Celsius over 50 degree Celsius ambient).

Short time current test

Corona (dry) and RIV (dry) test (for 132kV and above voltage level clamps)

Resistance test and tensile test

Cantilever Test

3.1.12 Enamel, if used for conductor insulation, shall be either polyvinyl acetate type or amide type and shall meet the requirements of IS:4800. Polyester enamel shall not be used. Double cotton cover, if used, shall be suitably covered to ensure that it does not come in contact with oil. The secondary winding insulation shall be of class 'H' type.

3.1.13 The temperature rise on any part of equipment shall not exceed the maximum temperature rise mentioned below under the conditions specified in test clauses. The permissible temperature rise indicated below is for a maximum ambient temperature of 50 deg.C.

Sl. No.	Nature of the part or of the liquid	Maximum value of	
		Temperature (deg.C)	Temp. rise at a max. ambient air temp. not exceeding 50 deg. C

		(deg.C)
1. Contacts in air, silver-faced copper, copper alloy or aluminium alloy (see notes (i) and (ii))	105	55
Bare copper or tinned aluminium alloy.	75	25
2. Contacts in oil: Silver-faced copper, copper alloy or aluminum alloy (see note ii)	90	40
Bare copper or tinned aluminum alloy	80	30
3. Terminals to be connected to external conductors by screws or bolts silver faced (see note iii)	105	55
Bare	90	40
4. Metal parts acting as springs	-----(See note iv)-----	
5. Metal parts in contact with insulation of the following classes:		
Class Y :(for non-impregnated materials)	90	40
Class A :(for materials immersed in oil or impregnated)	100	50
Class E : in air	120	70
in oil	100	50
Class B : in air	130	80
in oil	100	50
Class F : in air	155	105
in oil	100	50
Enamel : oil base	100	50
synthetic, air	120	70

Sl. No.	**Nature of the part or of the liquid**	**Maximum value of**	
		Temperature (deg.C)	**Temp. rise at a max. ambient air temp. not exceeding 50 deg. C (deg.C)**

6.	Any part of metal or of insulating material in contact with oil except contacts	100	50
7.	Oil	90	40

Note: i) When applying the temperature rise of 55 deg.C, care should be taken to ensure that no damage is caused to the surrounding insulating materials.

ii) The quality of the silver facing shall be such that a layer of silver remains at the point of contact after the mechanical endurance test. Otherwise, the contacts shall be regarded as "bare".

iii) The values of temperature and temperature rise are valid whether or not the conductor connected to the terminals is silver-faced.

iv) The temperature shall not reach a value where the elasticity of the material is impaired. For pure copper, this implies to a temperature limit of 75 deg.C.

3.1.14 The 220 kV and 132 kV Current Transformers shall be provided with suitable test tap for measurement of capacitance, tan delta as well as partial discharges, in factory as well as at site. Provision shall be made of a screw on cap for solid and secured earthing of the test tap connection, when not in use. A suitable caution plate shall be provided duly fixed on the cover of the secondary terminal box indicating the purpose of the test tap and necessity of its solid Earthing as per prescribed method before energizing the Instrument Transformer.

All instrument transformers should have sealing bolts with holes for providing seals on opposite sides of the main tank flange at top, bottom and secondary terminal box.

A Partial discharge Test clause 5.1.2.2, 8.2.2.2 of IEC 60044-1

a) Measurement at highest system voltage Um for which limiting value is 10 Pc
b) Measurement at 1.2 Um/√3 for which limiting value is 5 Pc.

B. Tan Delta Test Clause 5.1.2.4 of IEC 60044-1

Maximum Limiting value $5x10^{-3}$

Measuring value to be 5kV to 1.2 Um/√3(i.e at 5kV,10kV, 0.3 Um/√3,0.7 Um/√3, Um/√3,1.2 Um/√3)

4.0 CURRENT TRANSFORMERS

4.1 STANDARDS:

The current transformers and associated accessories shall conform to the latest issues/amendments of standards given below, except to the extent explicitly modified in the specification.

Sl. No.	Standard No.	Title
1.	IS:2165	Insulation Co-ordination for equipment of 100 kV and above.
2.	IS:2705	Current Transformers (I to IV)
3.	IS:2099	High voltage porcelain Bushings

4. IS:3347 Dimensions of porcelain transformer bushings.

5. IS:2071 Method of High Voltage Testing.

6. IS:335 Insulating oil for transformers and switch gears

7. IS:2147 Degree of protection provided by enclosures for low voltage switch gear and control.

8. IEC-185 Instrument Transformers.

9. IEC-270 Partial Discharge Measurement.

10. IEC-44(1) Current Transformer measurement of PDs & Tan Delta.

11. IEC-171 Insulation co-ordination.

12. IEC-60060 High voltage testing techniques

13. IEC-8263 Method for RIV test on high voltage insulators

14. IEC Indian Electricity Rules 1956.

15. IS:16227 /IEC 61869 Part 1 & 2 Latest IS for CT's

16. IEC 60060 Method of High Voltage Testing.

17. IEC 60529 Degree of protection provided by enclosures for low voltage Switch gear and control

18. IEC-60270 Partial Discharge Measurement

19. IEC- 60071 Insulation co-ordination

20. IEC 60437 Method for RIV test on high voltage insulators for RIV test on high voltage insulators

21. IEC 61462 Composite Insulators

Note: IS:16277 (section5.3.3.1) the 33KV CT's shall meet the partial discharge requirements applicable to instrument Transformers having Um greater than or equal to 7.2KV.

Reference Abbreviation	Name and Address
IEC	International Electro Technical Commission, Bureau central de la commission, Electro Technique international, 1, Rue de vereimbe Geneva. , Switzerland

IS Indian standard, Bureau of Indian Standards, Manak Bhavan,9, Bahadur shah Zafar Marg, New Delhi- 110 002.

BS British Standards,
British Standards Institution, 101, Pentonnvilla Road,

N -19 - ND - UK.

4.1.1 In case equipment conforms to other international standards which ensure equal or better performance than that specified under clause 2,then the English version of such standards or the relevant extracts of the same shall be furnished with the bid and the salient features of comparison shall be brought out separately in additional information schedule.

4.2. PRINCIPAL PARAMETERS:

The Current Transformers shall conform to the following specific parameters.

Sl. No.	Item	Specification		
		220 kV	**132 kV**	**33 kV**
1.	Type of Installation	Single phase, dead tank oil filled hermetically sealed and outdoor type		Single Phase conventional oil filled hermetiacally sealed/dry type(polycrete) outdoor type
2.	Type of mounting	---------- Pedestal Type ----------		
3.	Suitable for system Frequency	----------50 Hz+/- 5%----------		
4.	Highest System voltage (kV rms)	245	145	36
5.	Current Ratio A/A	See Annexure III		
6.	Ratio taps	----------On secondary side----------		
7.	Method of earthing	----------Effectively grounded neutral system---------		
8.	Rated continuous thermal current (A)	----------120% of rated current----------		
9.	Acceptable limit of temperature rise above the specified ambient temperature for continuous operation at rated current	----------As per IS:2705----------		
10.	Acceptable partial discharge level at 1.1 times the rated voltage	----- As per Technical Clause No.3.1.14 (A)-------		Applicable for dry type CT(Polycrete) only.
11.	1.2/50 microsecond lightning impulse withstand voltage (kVp)	1050	650	170
12.	1 minute dry power frequency withstand voltage for primary	460	275	70

	winding (kV rms)			
13.	1 minute dry Power frequency over-voltage withstand voltage for secondary winding (kV rms)	3	3	3

Sl. No.	Item	Specification 220 kV	132 kV	33 kV
14.	Min. total creepage distance of porcelain housing (mm)	7595	4495	1116
15.	Rated short time withstand current for 1 second duration (kA rms)	50	40	25 for 3 sec
16.	Rated dynamic withstand current (kAp)	125	100	62.5
17.	Type of Insulation	----------Class A----------		
18.	R.I.V at 1.1 X rated voltage	Less than 500 micro volts.		
19.	20% spare terminals shall be provided			

4.3 The current transformer shall be of dead tank/Live tank design for 220 kV ~~and~~ ,132 kV CTs and live tank for 33kV CTs, shall be so designed constructed that it can be easily transported to site within the allowable transport limitation in vertical position. If the design is Live tank, the primary shall be Bar type and for Dead tank only hair pin design is allowed.

For 132kV and above live tank current transformers, stainless steel metal bellows shall be provided for compensation of oil volume variation.

For 132kV & above dead tank CTs Nitrogen cushioning is allowed (measures taken to minimize contact of nitrogen with oil is to be clearly explained in the drawing).

4.4 For live tank CTs, for compensation of variation in the oil volume due to ambient variation stainless steel bellows shall be used for all voltage levels. Rubber diaphragms shall not be permitted for this purpose.

4.5 The current transformer secondary terminals shall be brought out in a weather proof terminal box. For terminating secondary connections from windings, molded oil-resistant Epoxy cast resin terminal cover should be used. “O Rings” should be as per clause 3.1.6.2, 3.1.6.3. The terminal box shall be provided with removable gland plate and glands suitable for 1100 volts grade, PVC insulated, PVC sheathed multi-core stranded copper conductor cable. The terminal box shall be dust and vermin proof. There should be separate terminal box for the dedicated metering core and should be provided with proper sealing. Suitable arrangement shall be made for drying of air inside the secondary terminal box.

The dimensions of the terminal box and its openings shall be adequate to enable easy access and working space with use of normal tools. In addition to CT secondary existing terminals on extra set of secondary terminals shall be brought out on top of the original terminals to avoid direct tapping of original terminals to avoid leakage of oil. 3Nos. 6Cx 2.5 Sq.mm copper control cable shall be provided for 145KV&33KV CTs & 5Nos. 6Cx 2.5 Sq.mm copper control cable shall be provided for 245kV CTs. Double compression cable glands shall be supplied by the CT manufacturer as required.

4.6. Polarity shall be indelibly marked on each primary and secondary terminal. Facility shall be provided for short circuiting and grounding of the current transformer secondary terminals inside the terminal box.

4.7 The current transformer shall be provided with a rating plate with dimensions and markings as per IS:2705. The markings shall be punched and not painted.

4.8. The current transformer shall be vacuum filled with oil after processing and thereafter hermetically sealed to eliminate breathing and to prevent air and moisture from entering the tanks. Oil filling and/or oil sampling cocks, if provided to facilitate factory processing, should be permanently sealed before dispatching the current transformer. The method adopted for hermetic sealing shall be described in the offer.

4.9. The castings of base, collar etc., shall be die cast and tested before assembly to detect cracks and voids if any.

4.10 The Instrument security factor of metering core shall be less than 5. This shall be demonstrated on all the ratios of the metering core, in accordance with procedure specified in IEC-185 or IS:2705.

4.11 Core : All the cores may be continuous without any air-gap. The area of cross-section of the core and flux density at rated primary current and rated burden shall be consistent with the required characteristics of CTs. The cores shall be carefully annealed and bonded after they are wound, to relieve the stress during winding.

All the protection cores shall be able to produce undisturbed secondary current under transient conditions at all ratios.

4.12 Winding : The main insulation for the winding shall consist of paper and oil system which will possess high mechanical strength, superior electrical withstand properties and good ageing qualities to ensure long trouble free life for the CTs . The windings shall have high mechanical strength to withstand short-circuit stresses. It is preferable that the manufacturers have an auto clave facility

4.13 Primary Winding:

For 33kV CTs: For 33kV CTs primary must be a single copper rod of circular cross section of Diameter 30 mm up to 800 amps/single 50 mm diameter rod for current rating above 800 Amps. The secondary connections from the secondary winding shall be of Teflon insulation. The insulators for primary studs/terminals shall be of FRP or Epoxy cast resin. The continuous and extended rating current density for primary should not exceed 1.65 Amps/Square mm. The short circuit current density should meet relevant standards. The Bidder shall furnish in his offer detailed calculations for selection of winding cross-sections.

For 132kV and above: Primary winding shall consist of Rigid primary of suitable design. Unavoidable joints in the Primary winding shall be of welded type. The winding shall be Hair pin

type for dead tank and Rigid bar type for Live tank. The conducting material for primary winding may consist of electrolytic high conductivity rigid copper rod or rigid Aluminum tube /flat of sufficient area of cross-section to cater for the guaranteed short time as well as continuous thermal current rating. Flexible copper/Aluminum conductor as primary is not allowed. The bidder may offer CTs having bar type primary with inverted arrangement of construction if considered more appropriate. The details of such welded joints shall be indicated in the drawings submitted with the offer. The design density for short circuit current as well as conductivity of the metal used for primary winding shall meet the requirement of IS 16227/IEC 61869. The continuous and extended rating current density for primary should not exceed 1.65 Amps/Sq. mm in case of copper & 1.0A/Square mm for Aluminium . The short circuit current density should meet relevant standards.

In case of Live tank CTs:

(i) The primary conductor shall preferably be of bar type meeting the desired Characteristics.

(ii) The secondaries shall be totally encased in rigid Aluminum metallic shielding providing a uniform equipotential surface for even electric field distribution.

(iii) The lowest part of the insulation assembly i.e. insulation at neck shall be properly secured to avoid any risk of damage due to transportation stresses.

(iv) The upper part of insulation assembly resting on primary bar shall be properly secured to avoid any damage during transportation due to relative movement between insulation assembly & top dome.

(iv) Bellows made of stainless steel shall be used at the top for hermetic sealing of CT.

4.14 Secondary Windings:

Suitably insulated copper wire of high conductivity electrolytic grade shall be used for secondary windings. Type of insulation used shall be described in the offer. For multi ratio design, suitable tappings shall be provided on the secondary winding only. Primary reconnection to achieve different ratios on secondary is not acceptable.

4.15 The exciting current of the current transformer shall be as low as possible. The Bidder shall furnish along with his offer the magnetization curve(s) for all the core(s).

4.16. Primary Terminals:

For 33kV CT: The dimensions should be as per clause 4.12. The surface of the terminal outside the tank shall be Nickel plated. The minimum thickness of tinning shall be 15 microns.

132kV and Above Voltage level: For ratings above 800 Amps, the primary terminals shall be of one number 40 mm Diameter copper rod or Aluminum pad of sufficient size. For ratings equal to and below 800 Amps, the primary terminals shall be of 30 mm Diameter copper rod or 40mm Aluminum pad/rod. The primary terminals shall be of heavily tinned electrolytic copper or aluminum alloy of 99.9% conductivity. The minimum thickness of tinning shall be 15 microns

4.17. Secondary Terminals:

Secondary terminal studs shall be provided with at least three nuts and adequate plain and spring washers for fixing the leads. The studs, nuts and washers shall be of brass, duly nickel plated. The minimum outside diameter of the studs shall be 6mm. A length of at least 15 mm shall be available on the studs for inserting the leads. The horizontal spacing between centers of the adjacent studs shall be at least 1.5 times the outside circum dia of the nuts.

For terminating secondary connections from windings, oil-resistant Epoxy molded cast resin terminal cover should be used. "O" Rings should be as per clause 3.1.6.3. The external wiring from cast resin terminal cover shall be wired to a separate terminal plate. The field cables should be terminated on this second terminal plate only.

4.18. Explosion Vent

Explosion vent shall be provided in expansion chamber for all class of CTs for safeguarding the equipment.

5.0 CAPACITOR VOLTAGE TRANSFORMERS :

5.1. STANDARDS:

Unless otherwise specified elsewhere in this specification the rating as well as performance and testing of the Capacitor voltage transformers and associated accessories shall confirm to the latest issues/amendments of standards available at the time of placement of order of all the relevant standards as listed hereunder.

Sl. No.	Standard No.	Title
	CAPACITOR VOLTAGE TRANSFORMERS	
1.	IS:3156 (PART-IV)	Capacitor Voltage Transformers
2.	IS:2099	High Voltage porcelain bushings
3.	IS:2071	Method of High Voltage Testing
4.	IS:335	Insulating oil for transformers and switchings.
5.	IS:2165	Insulation Co-ordination for equipment of 100 kV and above..
6.	IEC:186	Voltage Transformers :- Chapter –III Capacitor Voltage Transformers:- Chapter – IV
7.	IEC:186 A	First supplement to IEC Publication 186
8.	IEC-270	Partial Discharge Measurement..
9.	IEC-171	Insulation co-ordination.
10.	IEC-358	Coupling capacitor divider.
11.	IEC-60	High Voltage Testing Techniques.

12.	IS: 9348	Coupling capacitors and capacitor dividers.
13.	IEC61869 -1,61869-5	Instrument transformers
14.	IEC-60358	Coupling capacitors and capacitor dividers
15.	IS-3156 Part 1-4	Voltage Transformers

Reference Standards	Name and Address
IEC	International Electro Technical Commission, Bureau centraldela commission, Electro Technique International, 1, Rue de vereimbe Geneva. , Switzerland
ISO	International Organizations for Standardization Danish Board of Standardization, Aurehoegvaj – 12 DK – 2900, Heelprup DENMARK.
IS	Indian standard, Bureau of India Standards Manak Bhavan, Bahadur shah Zafar Marg, New Delhi- 110 002. INDIA.

5.1.1 Equipment meeting with the requirement of other authoritative standards which ensure equal or better performance than the standards mentioned above, shall also be considered .When the equipment offered by the Bidder conforms to other standards adopted and the standards specified in this specification shall be clearly brought out in the relevant schedule. Four copies of such standards with authentic translation in English shall be furnished along with the offer.

5.2 PRINCIPAL PARAMETERS:

The Capacitor Voltage Transformers shall conform to the following specific parameters.

Sl. No.	Item	Specification 132 kV	220 kV
	CAPACITOR VOLTAGE TRANSFORMERS:		
1.	Type of Installation	Single phase, oil filled hermetically sealed and outdoor type	Single phase, oil filled hermetically sealed and outdoor type
2.	Type of mounting	Pedestal Type	Pedestal Type
3.	Suitable for system Frequency	50 Hz +/- 5%	50 Hz +/- 5%
4.	Highest System voltage (kV rms)	145	245
5.	Voltage Ratio		
6.	(i) Rated primary Voltage (kV r.m.s)	132/√3	220/ √3
	ii)Secondary Voltage (Volts)		
	Winding – I (For Protection)	110/ √3	110/ √3
	Winding – II (For protection)	110 / √3	110/√3
	Winding – III (For Metering)	110/ √3	110 / √3
	(iii) Simultaneous output of Secondary (VA)		
	Winding - I	50 VA	50 VA
	Winding – II	50 VA	50 VA
	Winding - III	50 VA	50 VA

Sl. No.	Item	Specification 132 kV	220 kV
7.	Accuracy Class		
	Winding – I	3P	3P
	Winding – II	3P	3P
	Winding –III	0.2	0.2
8.	Method of earthing the system	Effectively earthed	Effectively earthed
9	1.2/50 microsecond lightning impulse withstand voltage(kvp)	650	1050
10	1 minute dry Power frequency wiithstand voltage for Primary winding (kV rms)	275	460
11	Min. creepage distance of porcelain housing (mm)	4495	7595
12	Rated Voltage factor	1.2 Continuous 1.5 for 30 seconds	1.2 continuous 1.5 for 30 seconds
13	Limits of voltage errors and Phase displacement	Accuracy class / Percentage Voltage error / Phase displacement in minutes	
		a) Metering 0.2: ± 0.2 / ± 10	
		b) Protection 3P: ± 3 / ± 120	
		c) Protection 3p: ± 3 / ± 120	
14.	Rated total capacitance (pf)	4400+10% - 5%	4400 +10% - 5%
15.	Acceptable limit of variation of total capacitance over the entire carrier frequency range	+ 50 % & –20 % of the rated capacitance	+ 50 % and – 20% of the rated capacitance
16.	Equivalent series resistance over the entire carrier frequency range (ohms)	<40	<40

17. Stray capacitance and stray conductance of low voltage terminal over the entire carrier frequency range	As per IEC- 350	As per IEC-350
18. Standard reference range of frequency for which accuracies are valid	96% to 102 % for protection 99% to 101% metering	96% to 102% for protection 99% to 101% metering
19. 1 minute power frequency withstand voltage :		
i) Low voltage terminal (HF) and earth terminal (kV r.m.s)	3	3
ii) Withstand voltage for secondary winding (kV r.m.s)	3	3
20. Partial discharge level at rated voltage for capacitance divider (pico coulombs)	< 10	< 10
21. Maximum Temperature rise over ambient of 50 deg.C	As per IEC – 186	As per IEC – 186
22. Rated total thermal burden (Simultaneous)	150 VA (3P) 50 VA(0.2)	150 VA(3P) 50VA (0.2)
23. Seismic acceleration (Horizontal)	0.3 g	0.3 g
24. Type of insulation	Class A	Class A

5.3 The Capacitor voltage transformer is required for protection , metering, telemetering and telecontrol purposes.

5.4 The capacitor voltage divider of the capacitor voltage transformer shall be mounted in shedded, oil filled porcelain bushing. The electro-magnetic unit consisting compensating reactor, intermediate transformer, protective and damping devices etc., shall be mounted in a hermetically sealed metal enclosure, which shall also be used as a mechanical support for the capacitor voltage divider. The terminals of the secondary windings of the electromagnetic units shall be brought out in a separate, weather proof terminal box, which shall be properly mounted on the wall of the metal enclosure of the electromagnetic unit. The terminal box shall be dust and vermin proof. Suitable arrangement shall be provided for drying of air inside the terminal box

5.5. The capacitor voltage divider shall capable of being used over a wide carrier frequency bandwidth. The value of capacitance shall be suitably chosen for reliable wide band carrier coupling and shall be as per the values indicated in clause 5.1. The reactance of the teed branch of the CVT shall be adjusted to minimize carrier loss. With carrier frequency coupling device connected to the earth lead of the intermediate coupling capacitor, the accuracy of the electromagnetic voltage transformer unit shall remain within the specified limits.

5.6 Volumetric variations of the insulating oil inside the capacitor voltage divider unit shall be suitably compensated. The arrangement provided for this purpose shall be clearly described in the offer. Each unit of the capacitor voltage divider shall be hermetically sealed.

5.7 The oil volume variation in metal enclosure of the electromagnetic unit shall be counter balanced by a suitably dimensioned gas cushion.

5.8 Electromagnetic Unit / Base Housing :

5.8.1 Steel enclosure housing the Electromagnetic unit shall be coated with atleast two coats of zinc rich epoxy painting which will also serve as a mounting base for the Capacitor stack and a housing for carrier accessories.

5.9 PRIMARY WINDING:

The primary winding shall be of suitably insulated electrolytic copper wire. Type of insulation used shall be described in the offer. A carrier frequency choke shall be provided in series with the primary winding to prevent the loss of signal.

5.10 SECONDARY WINDING:

Suitably insulated copper wire of electrolytic grade shall be used for secondary windings. Type of insulation used shall be described in the offer. Each core of secondary winding shall be protected by two sets of HRC fuses in parallel.

5.11 The Secondary terminals shall be brought out in a weather proof terminal box. The HRC fuses meant for protection of secondary winding shall also be located in the terminal box. The terminal box shall be provided with removable gland plate and glands suitable for 1100 voltage grade, PVC insulated, PVC sheathed multicore stranded copper conductor cable. The dimensions of the terminal box and its openings shall be adequate to enable easy access and sufficient working space for use of normal tools. Polarity shall be indelibly marked on each of the secondary terminals in the terminal box.

5.12 SECONDARY TERMINALS:

Secondary winding terminal studs shall be provided with at least three nuts, plain and spring washers for fixing the leads. The studs, nuts and washers shall be of brass, duly nickel plated. The minimum outside diameter of the studs shall be 6mm. A length of at least 15 mm shall be available on the studs for inserting the leads.

5.13. The capacitor voltage transformer shall be provided with a rating plate with dimensions and markings as per IS : 3156 (Part- IV). The markings shall be punched and not painted. Rating Plate shall include Name of Purchaser, warranty period, P.O.No., Creepage Distance and IEC reference standard.

5.14 **Insulating Oil /Gas**

Insulating oil to be used for Capacitor Voltage Transformers shall be of EHV grade and shall conform to IS : 335.
The SF6 gas shall comply with IEC-60376,60376A& 60376B and shall be suitable for use in the switchgear under operating condition.

6.0. POTENTIAL TRANSFORMERS :

6.1. PRINCIPAL PARAMETERS :

The Voltage Transformers shall confirm to the following specific parameters

Sl. No.	Item		Specification 220kV	132kV	33kV
a)	Type	:	Outdoor, Single phase, oil immersed and self cooled.		
b)	Mounting of tank	:	-- Bottom tank --		
c)	Nominal system voltage	:	220kV	132kV	33kV
d)	Highest system voltage	:	245kV	145kV	36kV
e)	Earthing of system	:	Effectively grounded neutral system		
f)	Frequency	:		-- 50 C/S –	
g)	Insulation level (primary)	:	460 kVrms	275 kVrms	70 kV rms
h)	One minute power frequency withstand test voltage	:		-- 3 kV rms -- (Secondary)	
i)	Impulse withstand voltage(peak)	:	1050 kV	650 kV	170kV
j)	Minimum creepage distance of	:	7595 mm	4495 mm	1116 mm
i)	Bottom-most portion of the bushing: at a height of (the structure design to meet this requirement shall be given by the tenderer		2500 mm	2500mm	
k)	i) Primary and secondary core details:		The VTs are intended for use on 3 phase system. These shall be of single phase with primary suitable for phase to earth connection. The primary voltage is 220/√3kV, 132/√3kVand 33/√3kV. The secondary core details are as indicated below:		

		--220 kV & 132 kV --			33 kV	
Secondary winding	:	I	II	III	I	II

Secondary voltage	:	110/√3	110/√3	110/√3	110/√3	110/√3
Class of accuracy	:	3P	3P	0.2	0.2	0.2
Rated burden (simultaneous)VA:		50	50	50	50	10

6.2 The PTs covered by this specification shall comply with the latest issue of ISS : 3156 or British Standard : 3941 or IEC – 186 except where specified otherwise.

The bushing shall be of single piece. PTs with cores mounted at the top or middle of the bushing are not acceptable. Bushings with metallic joints in the middle are also not acceptable. The oil used for the PTs shall conform to the requirements of IS-335 of latest issue. The PTs shall be supplied with first filling of oil.
The bushings of PTs shall conform to the ISS-2039 of latest issue.

7 TESTS:

7.1 Type Tests for CTs:

CTs of each voltage class offered in the bid shall be fully type tested as per the relevant standards. The type tests must have been conducted not earlier than five years form the date of bid opening. The Bidder shall furnish one set of the following type test reports as per IS-2705 along with offer.

I. Short time current test.
II. Temperature –rise test.
III. Lightning impulse test.
IV. High voltage power frequency wet withstand voltage test.
V. Determination of errors.
VI. Tests for Accuracy
VII. Electronic magnetic Compatibility tests
VIII. Verification of the degree of protection by enclosures
IX. Enclosure tightness test at ambient temperature
X. Pressure test for the enclosure

In addition to type test reports, the supplier has to furnish test reports for the following Tests -

XI. Chopped Impulse Test
XII. Multiple Chopped Impulse Test (245kV and above)
XIII. RIV Test (245kV and above)
XIV. Mechanical Test on Terminals (132kV and above with minimum cantilever load of 350 kg)
XV. Seismic Test (132kV and above)
XVI. Tan Delta Test at full Voltage
XVII. Thermal Stability Test
XVIII. Temperature Coefficient Test
XIX. Partial Discharge measurement.
XX. Corrosion test for 132kV and above.

7.1.1 Routine tests

Routine/Acceptance tests shall be conducted at the manufacturer's works. The manufacturer's testing lab should have NABL accreditation. All the equipment used for testing should have valid calibration at the date of testing.
All routine tests as per **IS 16227/IEC 61869** are to be carried out by the manufacturer.

7.1.2 Acceptance Tests:

In addition to the routine tests carried out by the manufacturer, the following acceptance tests are tobe carried out as per **IS 16227/IEC 61869** on **10% of the offered lot** in the presence of APTransco without any additional cost to APTRANSCO.
The following tests are to be conducted on the current Transformers.
A. Partial discharge Test clause 7.3.2 of IEC 61869- 1 with latest amendments.
a) Measurement at highest system voltage Um for which limiting value is 10 Pc
b) Measurement at 1.2 Um/√3 for which limiting value is 5 Pc. As per 5.3.3.1 of IEC 61869-1
c) Partial discharge measurement at 80% (PFVW) of the pre-stress voltage shall not exceed 100pC.
B. Measurement of Tan Delta and capacitance Test Clause 7.4.3 of IEC 61869-1 (for 132kV &above)
a) Tan delta Maximum Limiting value 4x10-3. Measuring value to be at 0.3, 0.7, 1.0 and 1.2Um/√3 and at 10 kV. The percentage variation of tan delta value from 10kV to Um/√3 should not exceed 15%. The value of tan delta at 10kV measured at factory will be reference value.
b) The tan-delta value shall not exceed 0.7% in the warranty period and in any variation during warranty period the CT shall be replaced by the manufacturer.
c) Measurement of Tan delta as a function of temperature (at ambient and between 80 °C and 90°C) and voltage (at 0.3, 0.7, 1.0, and 1.2 Um/√3). The tan delta shall not exceed 0.4%.
D. Power frequency voltage with stand tests on primary terminals.
E. Power frequency voltage with stand tests between sections, if applicable.
F. Power frequency voltage with stand tests on secondary terminals.
G. Tests on accuracy.
H Test for winding resistance, rated knee point E.M.F and exciting current at rated knee point E.M.F for all ratios.
I. Inter-turn over voltage test.
J. Verification of terminal marking and polarity.
K. Determination of the instrument security factor (FS) of measuring current transformer at all ratios.
L. Lighning Impulse voltage withstand test on primary terminals on one sample per order as per sec7.2.3 of IS 16227 part-1 2016, limited to three positive impulses followed by three negative impulses.
This test can be carried out at the manufacturer's premises. If the manufacturer does not have the facility, this test can be conducted at any third party NABL accredited laboratory in the presence of APTRANSCO. (This test is applicable for 145kV & 220kV CTs.) - on any one unit/offered lot.
M. SEALING TEST AND PROCEDURE: (For all voltage class CTs)
i. Test shall be performed on completely assembled unit.
ii. Test shall be performed during acceptance test on one randomly selected unit.
iii.Temperature of CT/PT/CVT under test will be elevated and maintained at 50(°C) and simultaneously it shall be subjected to internal pressure of 103 kPa (@1.1 kg/sq.cm) for 12 hrs.
iv. Arrangement shall be made by manufacturer to maintain required pressure and temperature for 12hrs.
v. During and after the test, there shall not be any oil leakage from any part or joint of CT.
vi. Readings of temperature, internal pressure applied and duration test along with observation of leakage, if any, shall be noted in inspection report.

7.2. Type Test for CVTs :

CVTs of each voltage class offered in the bid shall be fully type tested as per the relevant standards. The type tests must have been conducted not earlier than five years form the date of bid opening. The Bidder shall furnish one set of the following type test reports as per IEC: 186/ IS-3156 along with offer.

i. Temperature –rise test.
ii. Lightning impulse test.
iii. Ferro resonance test
iv. Transient response test
v. Verification of accuracy.
vi. Wet test for outdoor type transformers
vii. Enclosure tightness test at ambient temperature.
viii. Capacitance and tanδ measurement at power-frequency.
ix. Short-circuit withstand capability test.
x. Partial Discharge Voltage
xi. Type test for carrier frequency accessories.
xii. Visual Corona test for tests above 150 kV.
xiii. Natural frequency of Capacitor unit, determination of Resonant Frequency of Capacitor unit
xiv. Siesmic withstand test
xv. High frequency capacitance and equivalent series resistance measurement (as per IEC-358)
xvi. Electromagnetic compatibility tests.
xvii. Verification of the degree of protection by enclosures.
xviii. Pressure test for the enclosure.

7.2.1 Acceptance and Routine Tests:

All acceptance and routine tests for Instrument Transformers as stipulated in the relevant standards and partial discharge test and tan delta tests shall be carried out as per Technical clause No.3.1.14 (A) and (B), mechanical loading test on primary terminals shall be carried out compulsorily by the Bidder in the presence of Purchaser's representative. Capacitance & Tan delta shall be measured before and after HV withstand test.

The Routine tests as per IEC 61895-Part 1 & 5 & IEC 60358

1. Power-frequency voltage withstand tests on primary terminals.
2. Power frequency voltage withstand tests between sections (If applicable).
3. Power frequency voltage withstand tests on secondary terminals.
4. Partial discharge measurement.
5. Test for accuracy.
6. Enclosure tightness test at ambient temperature.
7. Pressure test for enclosure.
8. Ferro-resonance check.
9. Routine tests for carrier frequency accessories.

7.3 Type tests for PTs:

PTs of each voltage class offered in the bid shall be fully type tested as per the relevant standards. The type tests must have been conducted not earlier than five years form the date of bid opening. The Bidder shall furnish one set of the following type test reports as per IS-3156 along with offer.

i. Temperature –rise test.
ii. Lightning impulse test.
iii. High voltage power frequency wet withstand voltage test.
iv. Determination of errors and other characteristics.
v. Electromagnetic compatibility tests. (Not applicable for 33kV PTs)
vi. Verification of degree of protection by enclosures.
vii. Short-circuit withstand capability test.
viii. Seismic withstand test. (Not applicable for 33KV PTs)
ix. Determination of temperature coefficient test (Not applicable for 132kV &33kV)
x. Mechanical loading tests (Not applicable for 132kV & 33kV)

7.4 Acceptance and Routine Tests:

7.5 All acceptance and routine tests for Instrument Transformers as per the latest IEC61869-1 & 3/IS-16227 Part 1 & 3 and as stipulated in the relevant Technical clause No.3.1.14 (A) and (B), partial discharge test ,tan delta tests and oil leakage test shall be carried out by the Bidder in the presence of Purchaser's representative.

7.6. Immediately after finalization of the program of acceptance/routine testing, the Bidder shall give sufficient advance intimation to the Purchaser, to enable him to depute his representative for witnessing the tests.

7.7 **Routine tests:**

Routine/Acceptance tests shall be conducted at the manufacturer's works. The manufacturer's testing lab should have NABL accreditation. All the equipment used for testing should have valid calibration at the date of testing.

8. INSPECTION:

i) The Purchaser shall have access at all times to the works and all other places of manufacture, where the Instrument Transformers are being manufactured and the Bidder shall provide Purchaser representative all facilities for unrestricted inspection of the works, raw materials, manufacture of all the accessories and for conducting necessary tests as detailed herein.

ii) The Bidder shall keep the Purchaser informed in advance of the time of starting and of the progress of manufacture of equipment in its various stages so that arrangements could be made for inspection.

iii) No equipment shall be dispatched from its place of manufacture unless the equipment has been satisfactorily inspected and tested.

iv) The acceptance of any quantity of the equipment shall in no way relieve the Bidder of his responsibility for meeting all the requirements of this specification and shall not prevent subsequent rejection if such equipment is later found to be defective.

The bidder shall give 15 days (for local supplies)/ 60 days (in case of foreign supplies) advance, intimation to enable the purchaser to depute his representative for witnessing acceptance and routine tests. All charges in connection with inspection by Purchaser's personnel such as travel, accommodation and incidentals shall be borne by the purchaser.

v) The Purchaser reserves the right to insist for witnessing the acceptance/routine testing of the bought out items. (The firm shall inform the APTRANSCO before procuring bought out items at the time of approval of drawings.)

vi) "APTRANSCO Engineer will not inspect incomplete materials, If APTRANSCO Engineer returned without inspections of offered equipments, due to Non-Readiness of materials at their works, the financial loss to the APTRANSCO, for deputing Engineer representative will be recovered from the concerned supplier".

vii) If the material is not ready for inspection, after the requested date, when the assigned inspector enquiries over phone, the call will be rejected and the supplier has to raise a fresh inspection call.

viii) APTransco shall be under no obligation to inspect materials within the mandatory advance notice period of fifteen days.

9 QUALITY ASSURANCE PLAN: Refer Clause-3 of Section-I, General

10. DOCUMENTATION:

10.1 All drawings shall conform to International Standards Organization (ISO) 'A' series of drawing sheet/Indian Standards Specification. All drawings shall be in ink and suitable for micro filming. All dimensions and data shall be in S.I. Units only.

10.2 List of Drawings and Documents:

The Bidder shall furnish four sets of following drawings along with his offer:

a) General outline and assembly drawings of the equipment.

b) Graphs showing the performance of equipment in regard to magnetization characteristics.

c) Sectional views showing:

i) General Constructional Features.

ii) Materials/Gaskets/Sealing used.

iii) The insulation of the winding arrangements, method of connection of the primary/secondary winding to the primary/secondary terminals etc.,

iv) Porcelain/Polymer used and its dimensions along with the mechanical and electrical characteristics.

d) Arrangement of terminals and details of connection studs provided.

e) Name plate. (Shall include warranty period)

f) Schematic drawing.

g) Type test reports of equipment offered.

h) Test reports, literature, pamphlets of the bought out items and raw material.

10.3 The Bidder shall within 2 weeks of placement of order, submit four sets of final versions of all the above drawings for Purchaser's approval. The Purchaser shall communicate his comments/approval on the drawings to the Bidder within reasonable time. The Bidder shall, if necessary, modify the drawings and resubmit four copies of the modified drawings for Purchaser's approval within two weeks from the date of Purchaser's comments.

After receipt of Purchaser's approval, the Bidder shall within three weeks, submit 20 prints and two good quality reproducible of the approved drawings for Purchaser's use.

10.4 Adequate copies of contract drawings, acceptance and routine test certificates, duly approved by the Purchaser shall accompany each of the dispatched consignment.

10.5 The manufacturing of the equipment shall be strictly in accordance with the approved drawings and no deviation shall be permitted without the written approval of the Purchaser. All manufacturing and fabrication work in connection with the equipment prior to the approval of the drawing shall be at the Bidder's risk only.

10.6 Twenty (20) copies of nicely printed and bound volumes of operation, maintenance and erection manuals in English language, for each type and rating of equipment supplied shall be submitted by the Bidder for distribution, prior to the dispatch of the equipment. The manual shall contain all the drawings and information required for erection, operation and maintenance of the transformers. The manual shall also contain a set of all the approved drawings, type, routine & acceptance Test reports etc.

10.7 Approval of drawings by the Purchaser shall not relieve the Bidder of his responsibility and liability for ensuring correctness and correct interpretation of the latest revision of applicable standards, rules and codes of practices. The equipment shall conform in all respects to high standards of engineering, design, workmanship and latest revisions of relevant standards at the time of ordering and the Purchaser shall have the power to reject any material which in his judgement is not in full accordance therewith.

11 PACKING AND FORWARDING: Refer Clause-4 of Section-I, General

12 QUANTITY AND DELIVERY REQUIREMENTS : Refer Clause-5 of Section-I, General

13 SUPERVISION SERVICES : Refer Clause-6 of Section-I, General

14. MARKING : Refer Clause-7 of Section-I, General

15.0 GUARANTEED TECHNICAL PARTICULARS:

The Guaranteed Technical particulars for the equipment being supplied shall be provided with the Bid as specified in this technical specification at Annexure-I**.** The Bids received without guaranteed technical particulars shall be treated as Non-responsive.

ANNEXURE-I

GUARANTEED TECHNICAL PARTICULARS FOR CURRENT TRANSFORMERS

(To be filled in by the Bidder separately for each type and voltage rating)

1. Type of tank/Installation
   (Only Live tank are allowed for 33kV CTs)
2. Type of mounting
3. Manufacturer's Name and address and Country of Manufacture
4. Conforming to standard
5. a) i) Primary and Secondary winding made out of (As per clause 4.12,4.13,4.14)

   ii) Primary windings – Design density for short circuit current –
   Conductivity of metal used –

   b) i) Area of cross section of primary winding
   ii) Area of cross section of secondary winding

   c) Material used for providing secondary terminals

   d) Rated burden
6. Rated primary voltage (kV rms)
7. Rated highest voltage (kV rms)
8. Rated frequency (Hz)
9. Rated primary current (A)
10. Rated secondary current (A)
11. Ratio taps (on secondary side only)
12. Type of insulation
13. Seismic acceleration (g)
14. RIV at 1.1 x Rated voltage (mv)
15. Tank material and Tank coating (As per clause 3.1.3 & 3.1.4)
16. Hardware exposed to atmosphere (As per clause 3.1.4)
17. Bolts, Nuts and Washers (As per clause 3.1.4)
18. Porcelain housing and it make (Single piece only as per clause 3.1.2)

19. Sealing (Nitrogen gas cushion/Metal bellow as per clause 3.1.7)

20. Instrument security factor (As per clause 4.10)

21. Whether test tap provided (As per clause 4.18)

22. Acceptable limit of temperature

23. Acceptable partial discharge level at 1.1 times the rated voltage

24. Rated short time withstand current for 1 sec. duration (kA rms)

25. Rated dynamic withstand current (kAp)

26. Rated continuous thermal current (pu) where pu = rated current

27. 1.2/50 micro-second impulse withstand voltage (kVp)

28. One minute power frequency withstand voltage (kV rms) of primary winding (Dry)

29. One minute power frequency withstand voltage of secondary winding (kV rms)

30. a) Minimum total creepage distance of insulator bushing (mm)

    b) Protected creepage of distance of bushing (mm)

31. Details of Cores Core No. I II III IV V

31.1 Current Ratios A/A -- -- --- -- --

31.2 Output burden (VA) -- -- --- -- --

31.3 Class of accuracy __ __ __ __ __

31.4 Accuracy limit factor __ __ __ __ __

31.5 Min. knee point voltage (kpv) __ __ __ __ __

31.6 Secondary resistance __ __ __ __ __
corrected to 75 deg.C

31.7 Max. exciting current at __ __ __ __ __
20% of knee point voltage
50% of knee point voltage
100% of knee point voltage

31.8 Application -- -- -- -- --

32. Weight of oil (Kg.)

33. Total Weight (Kg.)

34. Mounting details

35 Overall dimensions

36. Characteristics (whether graphs enclosed):

a. Ratio and phase angle curves

b. Magnetisation curves

c. Ratio correction factor curves

37. Core

a) Area of cross section
b) Flux density at rated primary current and rated burden.

Name of the firm .....

Signature of Bidder.....

Name & Designation & Seal ……..

Date .....

ANNEXURE-II

CORE WISE DETAILS OF CURRENT TRANSFORMERS

Sl. No.	Rating	Description	CT Ratio	Class	Purpose
1.	220kV CTs	220kV Feeder CTs, PTR CTs/ Bus Coupler CTs, (50KA, 1 sec, Creepage: 31mm/KV)	Core-I to IV: 1600-800-500/1A Core-V: 1600-800-500-300/1A	Core:1 to 4: PX Vk: 1600-800-500 V Rct: 8-4-2.5 Ohms, Imag: 25-50-80mA Core-5: 0.2S @ 10VA: ISF<=5 at all ratios	220kV Feeder Protection: M1 Distance Protection, M2 Distance Protection, Bus-bar Protection, Spare Core-5: Metering 220/132kV PTRs HV Protection: Differential Protection, HV O/L & E/L Protection, Bus-bar Protection, Spare & Metering
2.	132kV CTs	132kV Feeder CTs & 132/33kV PTR HV & 220/132 KV PTR LV Dual core CTs (40KA, 1 sec, Creepage: 31mm/KV)	Core-I & Core- II: 800-600-500/1A Core-III: 800-600-500-300/1A	Core-1: PX (0.5 @5VA) Vk = 800V-600V-500V Rct: 4 – 3 – 2.5 Ohms Imag: 37.5 – 50 – 60 mA Core-2: 5P20 @ 20VA Core-3: 0.2S @ 10VA, ISF ≤5 at all Ratios	132kV Feeder CTs 132kV Feeder Protection: Main Distance Protection, O/L&E/F protection & Metering & 132/33kV PTR HV CTs 132kV PTR HV Protection: Differential protection HV O/L & E/L Protection Metering & 220/132kV PTR LV CTs 220kV PTR HV Protection Differential protection HV O/L & E/L Protection

					Metering

Rating	Description	CT Ratio	Class	Purpose
33kV CTs	33KV Feeder CTs & Capacitor Bank CTs (25kA/3sec Creepage:31mm/kV)	400/1-1 A	Core 1 : 5P10 @ 15VA Core 2 : 0.2S @ 10VA, ISF≤ 5	Core-1: For Overload & Earth fault protection Core-2: For Metering
33kV CTs	For PTRs LV Side (25kA/3sec Creepage:31mm/kV)	2000-1200-800/ 1-1-1-1 A	Core 1 : PS, Vk =40 (Rct+10) @ all ratios, Imag 30 mA @ 2000/1A 40 mA @ 1200/1A 50mA @ 800/1A Core 2 : 5P10 @ 15VA Core 3 : 2000-1200-800/1A, 0.2S @ 10VA, ISF ≤ 5 at all Ratios Core 4 : 2000-1200-800/1A, 0.2S @ 5VA, ISF ≤ 5 at all ratios	Core-1: For differential protection for PTR LVs Core-2: For Overload & Earth fault protection Core-3 : For Panel Metering Core-4: Boundary Metering (To be provided in separate terminal box)

* The terminal boxes shall be perpendicular to the primary terminal P1 & P2.

Note:

i. Additional one set of secondary terminals shall be brought on top of the secondary terminals to avoid direct tapping of the terminals.

ii. The terminal connectors to be supplied along with the equipment shall be suitable for both horizontal and vertical take-offs of the conductor.

iii. The ratios for the secondary core of the CTs shall be confirmed along with the drawing approval.

8. ANNEXURE-III

GUARANTEED TECHNICAL PARTICULARS FOR CAPACITOR VOLTAGE TRANSFORMERS

(To be filled in by the Bidder separately for each type and voltage rating)

1. Type of tank/Installation

2. Type of mounting

3. Manufacturer's Name, Address and Country of Manufacture

4. Conforming to standard

5. Type of voltage transformer

6. Type of insulation used

7. Rated primary voltage (kV rms)

8. Highest system voltage (kV rms)

9. Rated Secondary Voltage (volts)	For winding-I	Winding-II	Winding-III
10. Rated Transformation ratio	---	---	---
11. Rated secondary Burden(VA)	---	---	---
12. Rated thermal burden(VA)	---	---	---
13. Accuracy at rated burden	---	---	---
14. Rated frequency (Hz)	---	---	---

15. Temp. rise of winding by resistance (deg.c)

16. Primary winding one minute power frequency withstand voltage kV (r.m.s)

17. Impulse withstand Voltage 1.2/50 micro Sec.full wave positive And negative impulse withstand voltage KV(peak)

18. One minute power frequency withstand voltage of secondary winding kV (r.m.s)

19. Maximum dynamic P.F.overvoltage kV (peak)

20. Maximum Radio interference voltage at 266 kV r.m.s (micro volts)

21. Corona inception and extinction voltage 9 kV(r.m.s)

22. Capacitance (+ 10% -5%)

23. Total creepage distance of bushing (mm)

24. Overall height(mm)

25. Total weight (mm)

26. Type test report (Report No.)

27. Min. recommended spacing between voltage transformers

28. Outline dimension drawing No. giving clearance for installation (Drg.No.)

29. Mounting flange dimensional details.

30. Rated voltage Factor

31. Partial discharge level at rated voltage for capacitance divider (pc)

32. Seismic acceleration (horizontal) (g)

33. No. of terminals in terminal box

34. Primary and secondary windings made out of (copper only)

35. Tank material & tank coating (as per clause 3.1.3, 3.1.4)

36. Hardware exposed to atmosphere (as per clause 3.1.4)

37. Bolts, nuts and washers (as per clause 3.1.4)

38. Porcelain housing and its make (singular piece only as per clause 3.1.2)

39. Sealing (nitrogen gas cushion/metal bellow as per clause 3.1.7)

40. Whether supply includes terminal connectors.

Name of the firm .....

Signature of Bidder.....

Name, Designation & Seal .....

Date .....

ANNEXURE – IV

WINDING WISE DETAILS OF CAPACITOR VOLTAGE TRANSFORMERS

Sl	No.of Windings	Winding No.	Secondary Voltage (V)	Application Class	Accuracy out put	Simultaneous burden (VA)
1.	220 kV CVTs					
	3	1	110/√3	Protection	3P	50
		2	110/√3	Protection	3P	50
	3	3	110/√3	Metering	0.2	50
2	132kV CVTs					
	3	1	110/√3	Protection	3P	50
		2	110/√3	Protection	3P	50
		3	~~110~~-110/√3	Metering	0.2	50
3.	Rated thermal (Simultaneous) burden for the above					150

ANNEXURE – V

SCHEDULE OF TECHNICAL & GUARANTEED PARTICULARS FOR PTs

(To be filled in by the tenderer separately for each voltage rating)

		220 KV	132KV	33Kv
1.	Type			
2.	Manufacturer's type /Designation			
3.	Mounting of tank (Bottom)			
4.	Conforming to Standard			
5.	System rated voltage/highest voltage (kv)			
6.	Rated primary voltage (kv)			
7.	Secondary winding details:			
a)	No.of secondary windings	Winding – I	Winding – II	
b)	Rated secondary voltage(v)			
c)	Rated burden (VA)			
d)	Class of accuracy			
8.	Temperature rise at 1.1 times rated voltage with rated burden (deg.C)			
9.	Rated voltage factor and time.			
10.	Temperature rise for item – 10 above (deg.C)			
11.	Impulse withstand test voltage (KV) peak.			
12.	One minute power frequency withstand test voltage on primary (KV rms)			
13.	One minute power frequency withstand test voltage on secondary (KV rms)			
14.	Porcelain Bushing (make)			
15.	Total creepage distance of the bushing (mm)			
16.	Protected creepage distance of the bushing (mm)			
17.	Quantity of insulating oil (Litres)			
18.	Weight of oil (Kg.)			
19.	Total weight including oil (Kg).			
20.	Mounting details			

21. Live parts to ground clearance

22. Whether V.T. is sealed construction with nitrogen at top.

23. Overall dimensions:

Height mm

Length mm

Breadth mm

24. Grade of oil.

25. Voltage Transformers shall carry 25% above the rated burden continuously without injury in respect of voltage Transformers.

Name of the firm .....

Signature of Bidder.....

Name, Designation & Seal .....

Date .....

ANNEXURE – VI

WINDING WISE DETAILS OF PT

Sl.No.	Windings	Application No. Class	Winding Accuracy Voltage (V) output	Secondary simultaneous Burden VA
1)	**220kV PTs**			
	110V/√3	Protection	3P	50
	110V/√3	Protection	3P	50
	110V/√3	Metering	0.2	50
2)	**132kV PTs**			
	110V/√3	Protection	3P	50
	110V/√3	Protection	3P	50
	110V/√3	Metering	0.2	50
3)	**220kV Metering PT**			
	110V/√3	Metering	0.2	10
4)	**132kV Metering PT**			
	110V/√3	Metering	0.2	10
	33kV Metering PT			
	110V/√3	Metering	0.2	10
6)	**33kV PTs**			
	110V/√3	Metering	0.2	50
	110V/√3	Metering	0.2	10

Note : The ratios for secondary core of the PTs shall be confirmed along with drawing approval.

As per latest IS:16227(section5.3.3.1) the 33KV CT's shall meet the partial discharge requirements applicable to instrument Transformers having Um greater than or equal to 7.2KV.

ACRONYMS

MVA	: Mega Volt Ampere
KV	: Kilo Volt Ampere
kV	: Kilo Volt
Cu. M	: Cubic metre
Deg. C	: Degrees Centigrade
mm.	: Millimetre
cm.	: Centimetre

dm.	: Decimetre
km.	: Kilometre
g.	: Grams
kg.	: Kilogram
V	: Volts
Hz	: Hertz
DC	: Direct Current
AC	: Alternate Current
ONAN	: Oil Natural Air Natural
ONAF	: Oil Natural Air Forced
HV	: High Voltage
IV	: Intermediary Voltage
Db	: Decibel
KA	: Kilo Amperes
sec.	: Seconds
Amps.	: Amperes
kVp.	: Kilo volts peak
kV. rms.	: Kilo volts root mean square
KW	: Kilo watt
Max.	: Maximum
Min.	: Minimum
UTS	: Ultimate Tensile Strength
QAP	: Quality Assurance Plan
ACSR	: Aluminium Conductor Steel Reinforced
AAAC	: All Aluminium Alloy Conductor
HP	: Horse Power
ppm.	: Parts Per Million
OLTC	: On Load Tap Changer
WTI	: Winding Temperature Indicator
OTI	: Oil Temperature Indicator
LOI	: Letter of Intent
CIF	: Cost including Insurance & Freight
CIP	: Carriage Insurance Paid
SF6	: Sulphur Hexa Flouride
TRV	: Transient Recovery Voltage
CTD	: Capacitor Tripping Device
ICU	: Individual Compressor Unit
NO	: Normally Open
NC	: Normally Close
sq.mm Al. Cable	: Square millimeter Aluminium cable
sq.mm Cu. Cable	: Square millimeter Copper cable
MS Flat	: Mild Steel Flat
C	: Close
O	: Open
HRC	: High Rupturing Capacity
PVC	: Poly Vinyl Chloride

p.f	: Power factor
CT	: Current transformer
NCT	: Neutral Current Transformer
TEFC	: Totally enclosed fan cooled
P.U.	: Per Unit
ISO	: Indian Standard Organization
CB	: Circuit Breaker
Auto	: Automatic
kg./Sq.mm	: Kilogram per square centimeter

A C R O N Y M S

MVA	Mega Volt Ampere
KVA	Kilo Volt Ampere
kV	Kilo Volt
Cu. m	Cubic metre
Deg. C	Degrees Centigrade
mm.	Millimetre
cm.	Centimetre
dm.	Decimetre
km.	Kilometre
g.	Grams
kg.	Kilogram
V	Volts
Hz	Hertz
DC	Direct Current
AC	Alternate Current
ONAN	Oil Natural Air Natural
ONAF	Oil Natural Air Forced
HV	High Voltage
IV	Intermediary Voltage
Db	Decibel
KA	Kilo Amperes
sec.	Seconds
Amps.	Amperes
kVp.	Kilo volts peak
kV. rms.	Kilo volts root mean square
KW	Kilo watt
Max.	Maximum
Min.	Minimum
UTS	Ultimate Tensile Strength
QAP	Quality Assurance Plan
ACSR	Aluminium Conductor Steel Reinforced
AAAC	All Aluminium Alloy Conductor
HP	Horse Power
ppm.	Parts Per Million

OLTC	On Load Tap Changer
WTI	Winding Temperature Indicator
OTI	Oil Temperature Indicator
Torr	Torr.
LOI	Letter of Intent
CIF	Cost including Insurance & Freight
CIP	Carriage Insurance Paid
SF6	Sulphur Hexa Flouride
TRV	Transient Recovery Voltage
CTD	Capacitor Tripping Device
ICU	Individual Compressor Unit
NO	Normally Open
NC	Normally Close
sq.mm Al. Cable	Square millimeter Aluminium cable
sq.mm Cu. Cable	Square millimeter Copper cable

MS Flat	Mild Steel Flat
C	Close
O	Open
HRC	High Rupturing Capacity
PVC	Poly Vinyl Chloride
p.f	Power factor
CT	Current transformer
NCT	Neutral Current Transformer
TEFC	Totally enclosed fan cooled
P.U.	Per Unit
ISO	Indian Standard Organization
CB	Circuit Breaker
Auto	Automatic
kg./Sq. mm	Kilogram per square millimeter